

8 Proofs Involving Sets

Use the methods introduced in this chapter to prove the following statements.

2. Prove that $\{6n : n \in \mathbb{Z}\} = \{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\}$.

Proof. First we show $\{6n : n \in \mathbb{Z}\} \subseteq \{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\}$. Suppose $a \in \{6n : n \in \mathbb{Z}\}$. This means $a = 6n$ for some integer n . Therefore $a = 2(3n)$ and $a = 3(2n)$. From $a = 2(3n)$, it follows that a is a multiple of 2, so $a \in \{2n : n \in \mathbb{Z}\}$. From $a = 3(2n)$, it follows that a is a multiple of 3, so $a \in \{3n : n \in \mathbb{Z}\}$. Thus by definition of intersection of two sets, we have $a \in \{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\}$. Therefore $\{6n : n \in \mathbb{Z}\} \subseteq \{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\}$.

Next we show $\{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\} \subseteq \{6n : n \in \mathbb{Z}\}$.

Suppose $a \in \{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\}$. By definition of intersection of two sets, this means $a \in \{2n : n \in \mathbb{Z}\}$ and $a \in \{3n : n \in \mathbb{Z}\}$. Since $a \in \{2n : n \in \mathbb{Z}\}$, we know $a = 2k$ for some $k \in \mathbb{Z}$. Thus a is even. Since $a \in \{3n : n \in \mathbb{Z}\}$, we know $a = 3l$ for some $l \in \mathbb{Z}$. Since 3 is odd, if l were odd then $3l$ would be odd; as $a = 3l$ is even, it follows that l is even. Then $l = 2j$ for some $j \in \mathbb{Z}$. So we have $a = 3l = 3(2j) = 6j$. Since $j \in \mathbb{Z}$, it follows that $a \in \{6n : n \in \mathbb{Z}\}$. Thus $\{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\} \subseteq \{6n : n \in \mathbb{Z}\}$.

Therefore, since $\{6n : n \in \mathbb{Z}\} \subseteq \{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\}$ and $\{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\} \subseteq \{6n : n \in \mathbb{Z}\}$, it follows that $\{6n : n \in \mathbb{Z}\} = \{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\}$. $\square$

4. If $m, n \in \mathbb{Z}$, then $\{x \in \mathbb{Z} : mn \mid x\} \subseteq \{x \in \mathbb{Z} : m \mid x\} \cap \{x \in \mathbb{Z} : n \mid x\}$.

Proof. Suppose $a \in \{x \in \mathbb{Z} : mn \mid x\}$. Then $a = mnk$ for some $k \in \mathbb{Z}$. So we have $a = m(nk)$ and $a = n(mk)$. By definition of divisibility, $m \mid a$ and $n \mid a$. Thus $a \in \{x \in \mathbb{Z} : m \mid x\}$ and $a \in \{x \in \mathbb{Z} : n \mid x\}$. By definition of intersection of two sets, we have $a \in \{x \in \mathbb{Z} : m \mid x\} \cap \{x \in \mathbb{Z} : n \mid x\}$. Therefore $\{x \in \mathbb{Z} : mn \mid x\} \subseteq \{x \in \mathbb{Z} : m \mid x\} \cap \{x \in \mathbb{Z} : n \mid x\}$. $\square$

6. Suppose A, B and C are sets. Prove that if $A \subseteq B$, then $A - C \subseteq B - C$.

Proof. We prove the contrapositive: if $A - C \not\subseteq B - C$, then $A \not\subseteq B$. Assume $A - C \not\subseteq B - C$. Then there exists an element $x \in A - C$ such that $x \notin B - C$. Since $x \in A - C$, we have $x \in A$ and $x \notin C$. Since $x \notin B - C$, it follows that $x \notin B$ or $x \in C$. Because $x \notin C$, it follows $x \notin B$. Since $x \in A$ and $x \notin B$, by definition of a subset, we have $A \not\subseteq B$. $\square$

8. If A, B and C are sets, then $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

Proof. Observe the following sequence of equalities.

$$\begin{aligned}
 A \cup (B \cap C) &= \{x : x \in A \vee x \in B \cap C\} && \text{(definition of } \cup) \\
 &= \{x : x \in A \vee (x \in B \wedge x \in C)\} && \text{(definition of } \cap) \\
 &= \{x : (x \in A \vee x \in B) \wedge (x \in A \vee x \in C)\} && \text{(distributive law)} \\
 &= \{x : x \in A \cup B \wedge x \in A \cup C\} && \text{(definition of } \cup) \\
 &= \{x : x \in (A \cup B) \cap (A \cup C)\} && \text{(definition of } \cap) \\
 &= (A \cup B) \cap (A \cup C) && \text{(definition of set-builder)}
 \end{aligned}$$

$\square$

10. If A and B are sets in a universal set U , then $\overline{A \cap B} = \overline{A} \cup \overline{B}$.

Proof. Observe the following sequence of equalities.

$$\begin{aligned}
 \overline{A \cap B} &= \{x \in U : x \notin A \cap B\} && \text{(definition of complement)} \\
 &= \{x \in U : \neg(x \in A \wedge x \in B)\} && \text{(definition of } \cap) \\
 &= \{x \in U : x \notin A \vee x \notin B\} && \text{(DeMorgan's law)} \\
 &= \{x \in U : x \in \overline{A} \vee x \in \overline{B}\} && \text{(definition of complement)} \\
 &= \{x \in U : x \in \overline{A} \cup \overline{B}\} && \text{(definition of } \cup) \\
 &= \overline{A} \cup \overline{B} && \text{(definition of set-builder)}
 \end{aligned}$$

$\square$

12. If A, B and C are sets, then $A - (B \cap C) = (A - B) \cup (A - C)$.

Proof. Observe the following sequence of equalities.

$$\begin{aligned}
A - (B \cap C) &= \{x : (x \in A) \wedge (x \notin B \cap C)\} && \text{(definition of difference)} \\
&= \{x : (x \in A) \wedge \neg(x \in B \cap C)\} && \text{(notation)} \\
&= \{x : (x \in A) \wedge \neg(x \in B \wedge x \in C)\} && \text{(definition of } \cap \text{)} \\
&= \{x : (x \in A) \wedge (\neg(x \in B) \vee \neg(x \in C))\} && \text{(DeMorgan's law)} \\
&= \{x : (x \in A) \wedge (x \notin B \vee x \notin C)\} && \text{(notation)} \\
&= \{x : (x \in A \wedge x \notin B) \vee (x \in A \wedge x \notin C)\} && \text{(distributive law)} \\
&= \{x : x \in A - B \vee x \in A - C\} && \text{(definition of difference)} \\
&= \{x : x \in (A - B) \cup (A - C)\} && \text{(definition of } \cup \text{)} \\
&= (A - B) \cup (A - C) && \text{(definition of set-builder)}
\end{aligned}$$

□

14. If A, B and C are sets, then $(A \cup B) - C = (A - C) \cup (B - C)$.

Proof. Observe the following sequence of equalities.

$$\begin{aligned}
(A \cup B) - C &= \{x : x \in (A \cup B) \wedge (x \notin C)\} && \text{(definition of difference)} \\
&= \{x : (x \in A \vee x \in B) \wedge (x \notin C)\} && \text{(definition of } \cup \text{)} \\
&= \{x : (x \in A \wedge x \notin C) \vee (x \in B \wedge x \notin C)\} && \text{(distributive law)} \\
&= \{x : x \in A - C \vee x \in B - C\} && \text{(definition of difference)} \\
&= \{x : x \in (A - C) \cup (B - C)\} && \text{(definition of } \cup \text{)} \\
&= (A - C) \cup (B - C) && \text{(definition of set-builder)}
\end{aligned}$$

□

16. If A, B and C are sets, then $A \times (B \cup C) = (A \times B) \cup (A \times C)$.

Proof. Observe the following sequence of equalities.

$$\begin{aligned}
A \times (B \cup C) &= \{(x, y) : x \in A \wedge y \in B \cup C\} && \text{(definition of } \times \text{)} \\
&= \{(x, y) : x \in A \wedge (y \in B \vee y \in C)\} && \text{(definition of } \cup \text{)} \\
&= \{(x, y) : (x \in A \wedge y \in B) \vee (x \in A \wedge y \in C)\} && \text{(distributive law)} \\
&= \{(x, y) : (x, y) \in A \times B \vee (x, y) \in A \times C\} && \text{(definition of } \times \text{)} \\
&= \{(x, y) : (x, y) \in (A \times B) \cup (A \times C)\} && \text{(definition of } \cup \text{)} \\
&= (A \times B) \cup (A \times C) && \text{(definition of set-builder)}
\end{aligned}$$

□

18. If A, B and C are sets, then $A \times (B - C) = (A \times B) - (A \times C)$.

Proof. Observe the following sequence of equalities.

$$\begin{aligned}
A \times (B - C) &= \{(x, y) : x \in A \wedge y \in B - C\} && \text{(definition of } \times \text{)} \\
&= \{(x, y) : x \in A \wedge (y \in B \wedge y \notin C)\} && \text{(definition of difference)} \\
&= \{(x, y) : (x \in A \wedge y \in B \wedge x \notin A) \vee (x \in A \wedge y \in B \wedge y \notin C)\} && \text{(first disjunct is false)} \\
&= \{(x, y) : (x \in A \wedge y \in B) \wedge (x \notin A \vee y \notin C)\} && \text{(distributive law)} \\
&= \{(x, y) : x \in A \wedge y \in B \wedge \neg(x \in A \wedge y \in C)\} && \text{(DeMorgan's law)} \\
&= \{(x, y) : (x, y) \in A \times B \wedge (x, y) \notin A \times C\} && \text{(definition of } \times \text{)} \\
&= \{(x, y) : (x, y) \in (A \times B) - (A \times C)\} && \text{(definition of difference)} \\
&= (A \times B) - (A \times C) && \text{(definition of set-builder)}
\end{aligned}$$

□

20. Prove that $\{9^n : n \in \mathbb{Q}\} = \{3^n : n \in \mathbb{Q}\}$.

Proof. First we prove $\{9^n : n \in \mathbb{Q}\} \subseteq \{3^n : n \in \mathbb{Q}\}$. Assume $a \in \{9^n : n \in \mathbb{Q}\}$. This means $a = 9^n$ for some $n \in \mathbb{Q}$. Thus $a = 9^n = (3^2)^n = 3^{2n}$. Since $2n \in \mathbb{Q}$, we have $a \in \{3^n : n \in \mathbb{Q}\}$. Therefore $\{9^n : n \in \mathbb{Q}\} \subseteq \{3^n : n \in \mathbb{Q}\}$.

Now we prove $\{3^n : n \in \mathbb{Q}\} \subseteq \{9^n : n \in \mathbb{Q}\}$. Assume $a \in \{3^n : n \in \mathbb{Q}\}$. This means $a = 3^n$ for some $n \in \mathbb{Q}$. Thus $a = 3^n = (9^{1/2})^n = 9^{n/2}$. Since $n/2 \in \mathbb{Q}$, we have $a \in \{9^n : n \in \mathbb{Q}\}$. Therefore $\{3^n : n \in \mathbb{Q}\} \subseteq \{9^n : n \in \mathbb{Q}\}$.

Therefore, since $\{9^n : n \in \mathbb{Q}\} \subseteq \{3^n : n \in \mathbb{Q}\}$ and $\{3^n : n \in \mathbb{Q}\} \subseteq \{9^n : n \in \mathbb{Q}\}$, it follows that $\{9^n : n \in \mathbb{Q}\} = \{3^n : n \in \mathbb{Q}\}$. □

22. Let A and B be sets. Prove that $A \subseteq B$ if and only if $A \cap B = A$.

Proof. First we prove if $A \subseteq B$, then $A \cap B = A$.

Suppose $A \subseteq B$. Then every element of A is also an element of B . Thus an element is in both A and B exactly when it is in A . Therefore $A \cap B = A$.

Conversely, we prove if $A \cap B = A$, then $A \subseteq B$.

Suppose $A \cap B = A$ and let $x \in A$. Since $A = A \cap B$, it follows that $x \in B$. Thus every element of A is also an element of B . Therefore $A \subseteq B$. □

24. Prove that $\bigcap_{x \in \mathbb{R}} [3 - x^2, 5 + x^2] = [3, 5]$.

Proof. First we prove that $[3, 5] \subseteq \bigcap_{x \in \mathbb{R}} [3 - x^2, 5 + x^2]$.

Take any $a \in [3, 5]$. Since $x^2 \geq 0$ for every $x \in \mathbb{R}$, it follows that $3 - x^2 \leq 3 \leq a$ and $a \leq 5 \leq 5 + x^2$. Thus $3 - x^2 \leq a \leq 5 + x^2$, so $a \in [3 - x^2, 5 + x^2]$ for every $x \in \mathbb{R}$. Therefore $[3, 5] \subseteq \bigcap_{x \in \mathbb{R}} [3 - x^2, 5 + x^2]$.

Conversely, we prove $\bigcap_{x \in \mathbb{R}} [3 - x^2, 5 + x^2] \subseteq [3, 5]$.

Since $0 \in \mathbb{R}$, one of the intervals in the intersection is $[3 - 0^2, 5 + 0^2] = [3, 5]$. Because an intersection must be contained in each set being intersection, we have $\bigcap_{x \in \mathbb{R}} [3 - x^2, 5 + x^2] \subseteq [3, 5]$. □

26. Prove $\{4k + 5 : k \in \mathbb{Z}\} = \{4k + 1 : k \in \mathbb{Z}\}$.

Proof. First we show $\{4k + 5 : k \in \mathbb{Z}\} \subseteq \{4k + 1 : k \in \mathbb{Z}\}$.

Assume $a \in \{4k + 5 : k \in \mathbb{Z}\}$. This means $a = 4k + 5$ for some $k \in \mathbb{Z}$. Thus $a = 4(k + 1) + 1$. Since $(k + 1) \in \mathbb{Z}$, we have $a \in \{4k + 1 : k \in \mathbb{Z}\}$. Therefore, $\{4k + 5 : k \in \mathbb{Z}\} \subseteq \{4k + 1 : k \in \mathbb{Z}\}$.

Now we prove $\{4k + 1 : k \in \mathbb{Z}\} \subseteq \{4k + 5 : k \in \mathbb{Z}\}$.

Assume $a \in \{4k + 1 : k \in \mathbb{Z}\}$. This means $a = 4k + 1$ for some $k \in \mathbb{Z}$. Thus $a = 4(k - 1) + 5$. Since $(k - 1) \in \mathbb{Z}$, we have $a \in \{4k + 5 : k \in \mathbb{Z}\}$. Therefore, $\{4k + 1 : k \in \mathbb{Z}\} \subseteq \{4k + 5 : k \in \mathbb{Z}\}$. □